Exploratory Data Analysis (EDA) Insights Report

Group: 19

Week 2 Deliverable – Exploratory Data Analysis, Dashboard, Insight Report

Campaign Objective: Improve enrollment outcomes from outreach campaigns by at least 15%

Dataset: Cleaned Outreach and Application Dataset

1. Introduction

This Exploratory Data Analysis (EDA) report evaluates outreach and application data to identify patterns, inefficiencies, and opportunities that can support Excellerate's campaign objective of improving enrollment outcomes by at least 15%. The analysis focuses on understanding connection success, caller effectiveness, timing patterns, geographic performance, and data quality factors that directly influence enrollment conversion.

All insights presented are derived from the cleaned dataset and are supported by charts attached separately, in line with Excellerate's reporting expectations.

2. Dataset Overview

The dataset used for this analysis contains 37,882 outreach records across 18 variables. No missing values or duplicate records were detected, indicating strong structural data quality. Most variables are categorical, with fields such as Applicant ID, Date Contacted, Phone Number, and Country.

Despite being structurally clean, the dataset reveals operational challenges including highly skewed call outcomes, uneven caller workloads, and significant contact data validity issues that affect outreach effectivenes

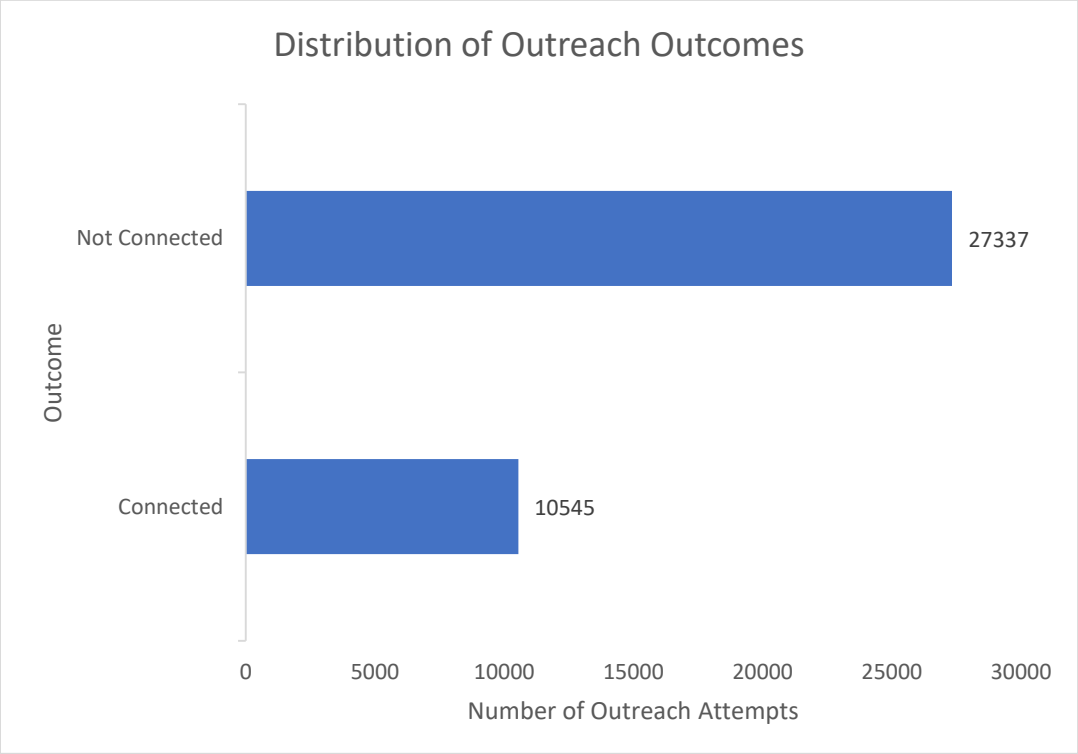
3. Key EDA Insights Aligned to the 15% Enrollment Improvement Goal

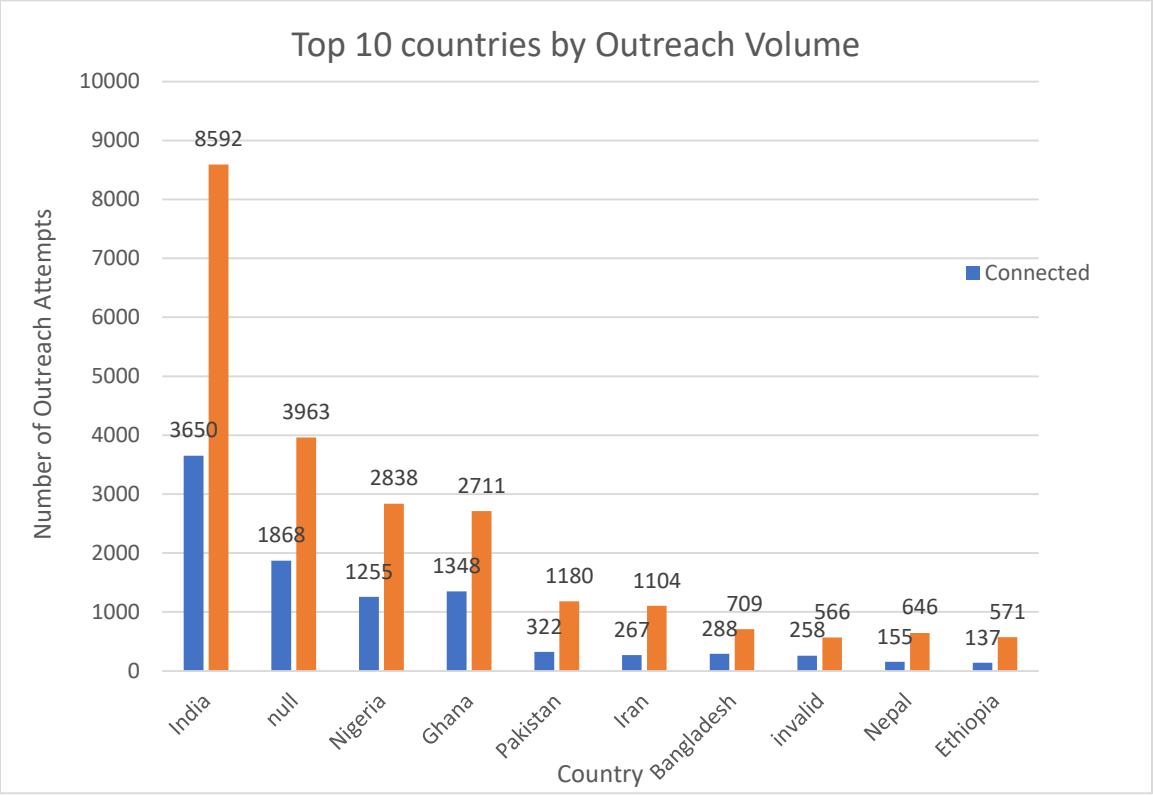
Insight 1: Low Connection Rates Are the Primary Barrier to Enrollment Growth

A significant majority of outreach attempts result in a “Not Connected” outcome, with unsuccessful calls occurring at more than six times the rate of successful connections. This indicates that most outreach effort does not translate into direct applicant engagement.

This is important metric because limited reachability directly constrains enrollment growth. Without a meaningful increase in the proportion of applicants successfully contacted, improvements in enrollment outcomes will remain difficult to achieve, regardless of how much outreach volume is increased.

Improving connection rates therefore represents the most impactful lever for driving enrollment growth. Enhancements in call timing, contact data quality, and engagement strategy that increase successful connections are likely to translate directly into higher enrollment outcomes and support the targeted 15% improvement.

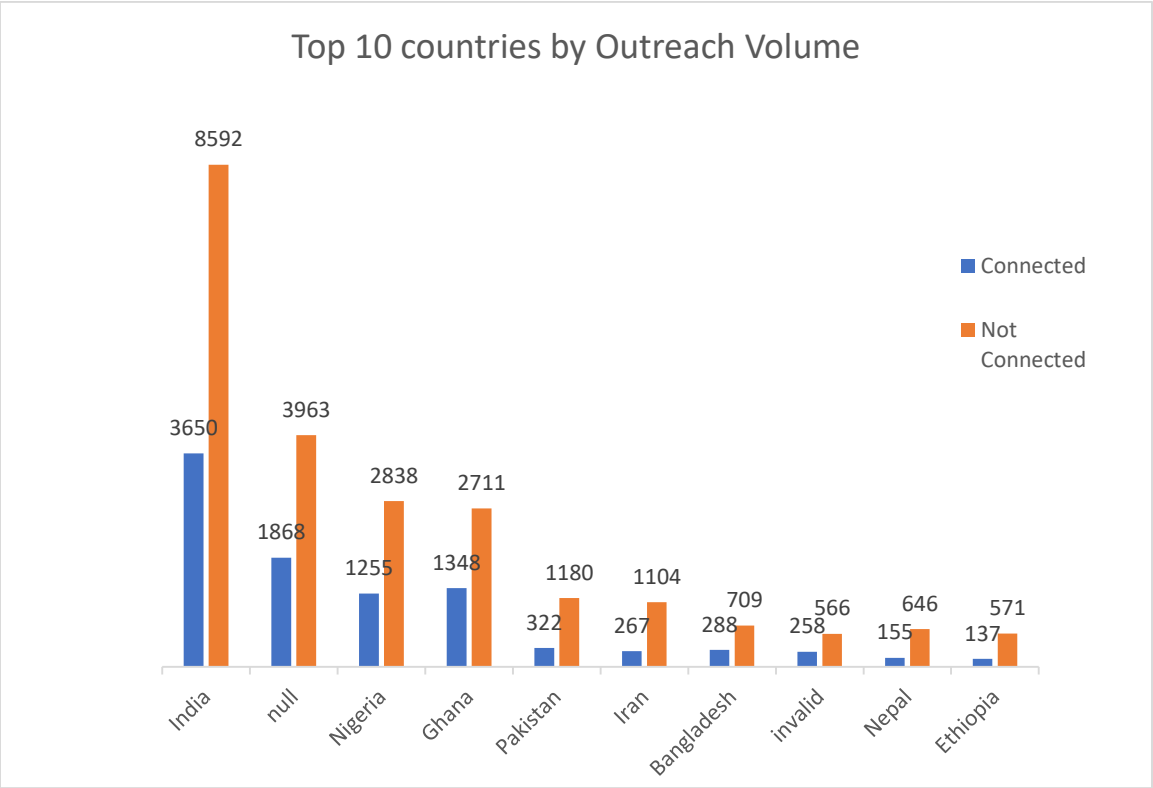



Top 10 countries by Outreach Volume
Number of Outreach Attempts
10000
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
Connected
3650
8592
1868
3963
1255
2838
1348
2711
322
1180
267
1104
288
709
258
566
155
646
137
571
India
null
Nigeria
Ghana
Pakistan
Iran
Bangladesh
invalid
Nepal
Ethiopia
Country

Insight 2: Enrollment and Outreach Performance Are Highly Concentrated by Country

Outreach activity is heavily concentrated in a few countries, with India accounting for the largest share by a wide margin. Across all top 10 countries, “Not Connected” outcomes substantially exceed successful connections, indicating that high outreach volume has not translated into proportional reachability. This highlights a reliance on scale rather than efficiency, which limits progress toward the 15% enrollment improvement goal.

High outreach volume does not equal high effectiveness. India shows the largest absolute gap between connected and not connected calls, while countries such as Ghana and Nigeria, despite lower volumes, demonstrate relatively better connectivity. Improving reachability through better timing and data validation especially in high-volume markets offers a more effective path to enrollment growth than increasing call volume alone.



Country	Connected
Bangladesh	28.89%
Ethiopia	19.35%
Ghana	33.21%
India	29.82%
invalid	31.31%
Iran	19.47%
Nepal	19.35%
Nigeria	30.66%
null	32.04%
Pakistan	21.44%
Grand Total	**29.44%**

Connectivity performance varies meaningfully across the top outreach countries, with rates ranging from 19.35% to 33.21% against a global average of 29.44%. Although India drives the highest outreach volume and enrollment, its connectivity rate (29.82%) is only marginally above average. In contrast, countries such as Ghana (33.21%) and Nigeria (30.66%) demonstrate stronger reachability despite lower outreach volumes.

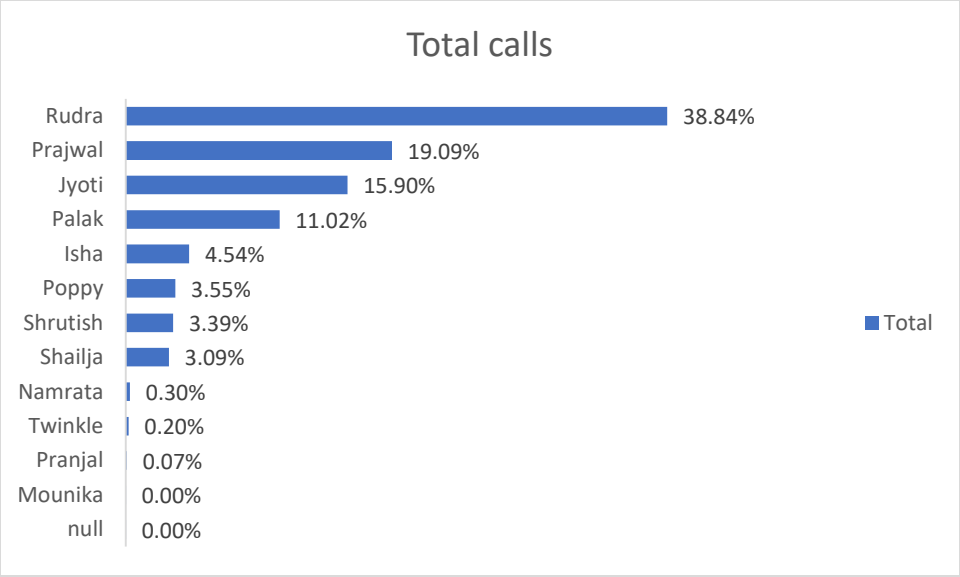
This indicates that current outreach efforts are weighted toward applicant volume rather than connectivity efficiency. To support the goal of improving enrollment outcomes by at least 15%, outreach should be optimized by prioritizing high-connectivity markets and improving reachability in high-volume regions through better call timing and contact data validation, rather than simply increasing call volume.

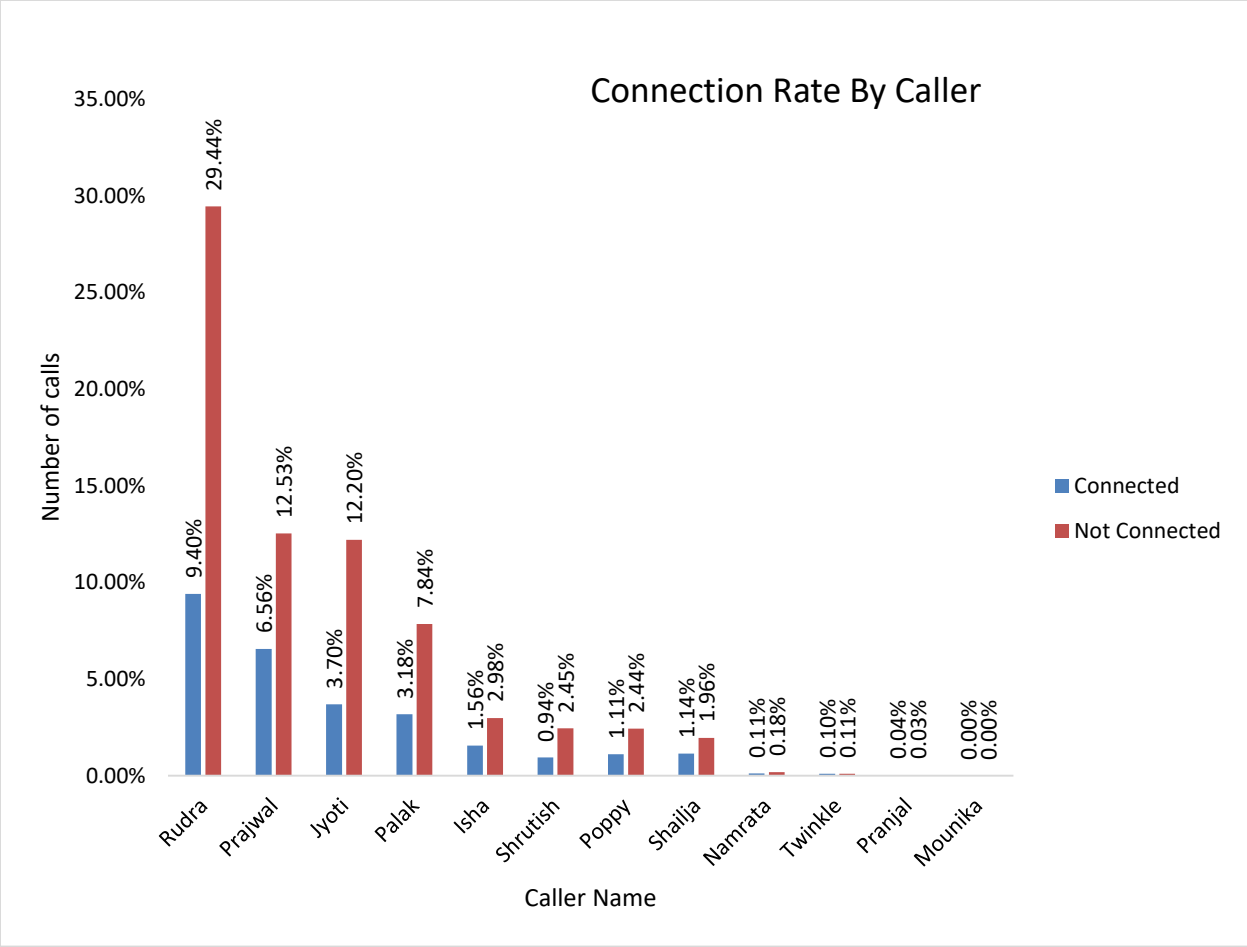
Focusing optimized outreach strategies on high-converting countries while redesigning approaches for underperforming regions can materially contribute to a 15% enrollment increase.

Insight 3: Caller Performance and Workload Distribution Affect Outreach Efficiency

Outreach volume is heavily concentrated among a small number of callers. Rudra alone accounts for approximately 39% of total calls**,** followed by Prajwal (19%) and **J**yoti (16%)**.** Together, these three callers contribute more than 70% of overall outreach activity, while the remaining callers each account for relatively small shares.

This concentration highlights a clear operational dependency on a few individuals to sustain outreach volume. While this does not by itself confirm overwork, it signals a potential risk to outreach consistency if performance, availability, or efficiency among these callers changes. Monitoring exposure levels for high-volume callers and aligning their outreach practices with team-wide standards will be important for maintaining stable performance and supporting the targeted 15% improvement in enrollment outcomes.


Total calls
Rudra 38.84%
Prajwal 19.09%
Jyoti 15.90%
Palak 11.02%
Isha 4.54%
Poppy 3.55%
Shrutish 3.39%
Shailja 3.09%
Namrata 0.30%
Twinkle 0.20%
Pranjal 0.07%
Mounika 0.00%
null 0.00%
Total


Connection Rate By Caller
Number of calls
35.00%
30.00%
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
9.40%
29.44%
6.56%
12.53%
3.70%
12.20%
3.18%
7.84%
1.56%
2.98%
0.94%
2.45%
1.11%
2.44%
1.14%
1.96%
0.11%
0.18%
0.10%
0.11%
0.04%
0.03%
0.00%
0.00%
Rudra
Prajwal
Jyoti
Palak
Isha
Shrutish
Poppy
Shailja
Namrata
Twinkle
Pranjal
Mounika
Caller Name
Connected
Not Connected

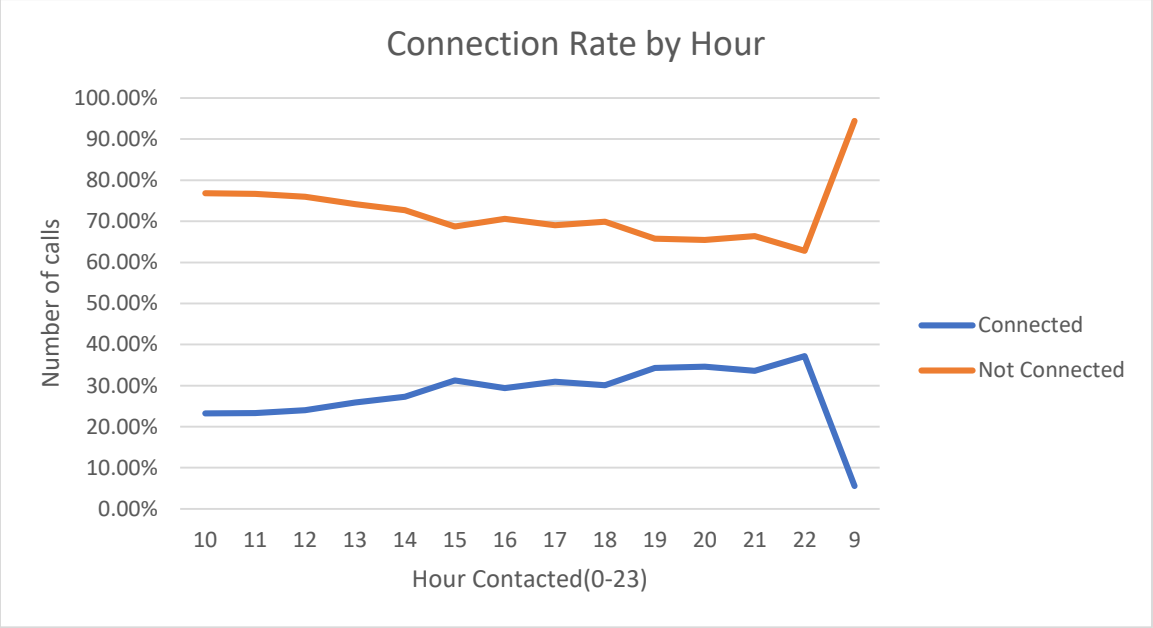
Insight 4: Time-of-Day Patterns Influence Connection Success

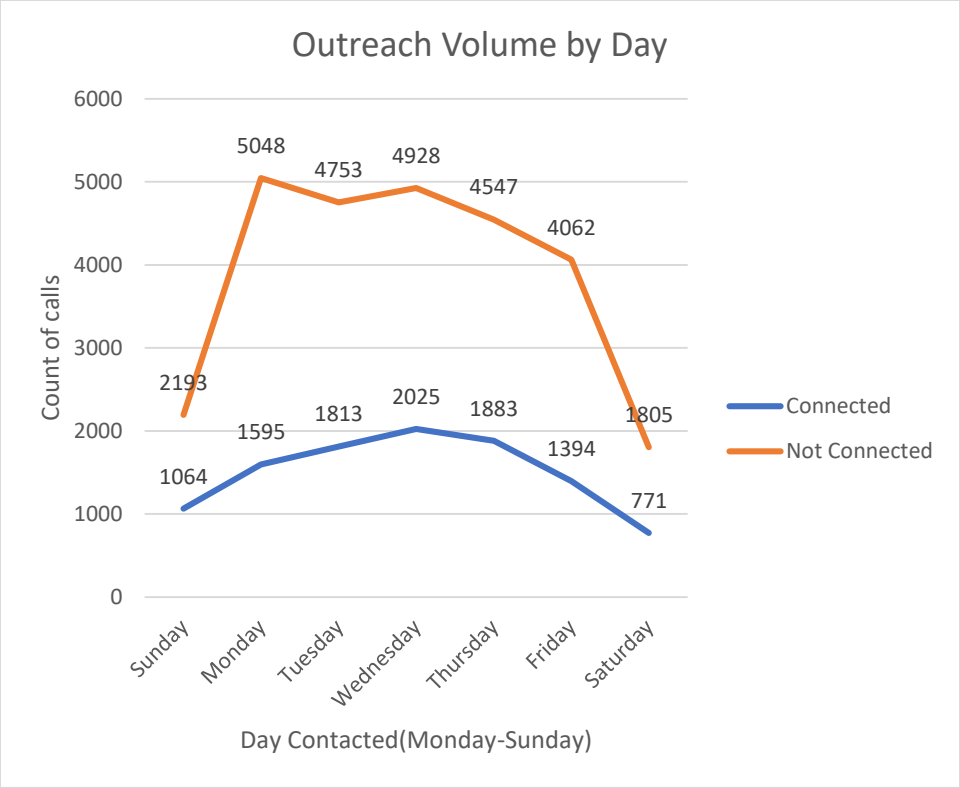
Most outreach attempts occur in the afternoon. However, evening calls, despite lower volume, demonstrate higher connection efficiency.

This chart shows connection outcomes by hour of contact, expressed as percentages. Each hour sums to 100%, split between *Connected* (blue) and *Not Connected* (red), allowing a fair comparison of effectiveness across time periods.

Connection effectiveness varies noticeably by hour of contact. During late morning and early afternoon (approximately 10:00–14:00), connection rates remain consistently low, with roughly three-quarters of calls resulting in no connection. As the day progresses into mid-afternoon and evening (15:00–22:00), connection rates steadily improve, reaching their highest levels in the evening hours. In contrast, early morning outreach around 09:00 performs particularly poorly, with very low connection success and a high proportion of unsuccessful calls.

This pattern indicates that applicant availability is strongly time-dependent. Concentrating outreach efforts in higher-performing evening windows, while reducing activity during early morning and low-yield daytime hours, provides a practical opportunity to improve overall reachability. Such timing optimization can increase the number of successful connections without increasing call volume, directly supporting the goal of improving enrollment outcomes by at least 15%.





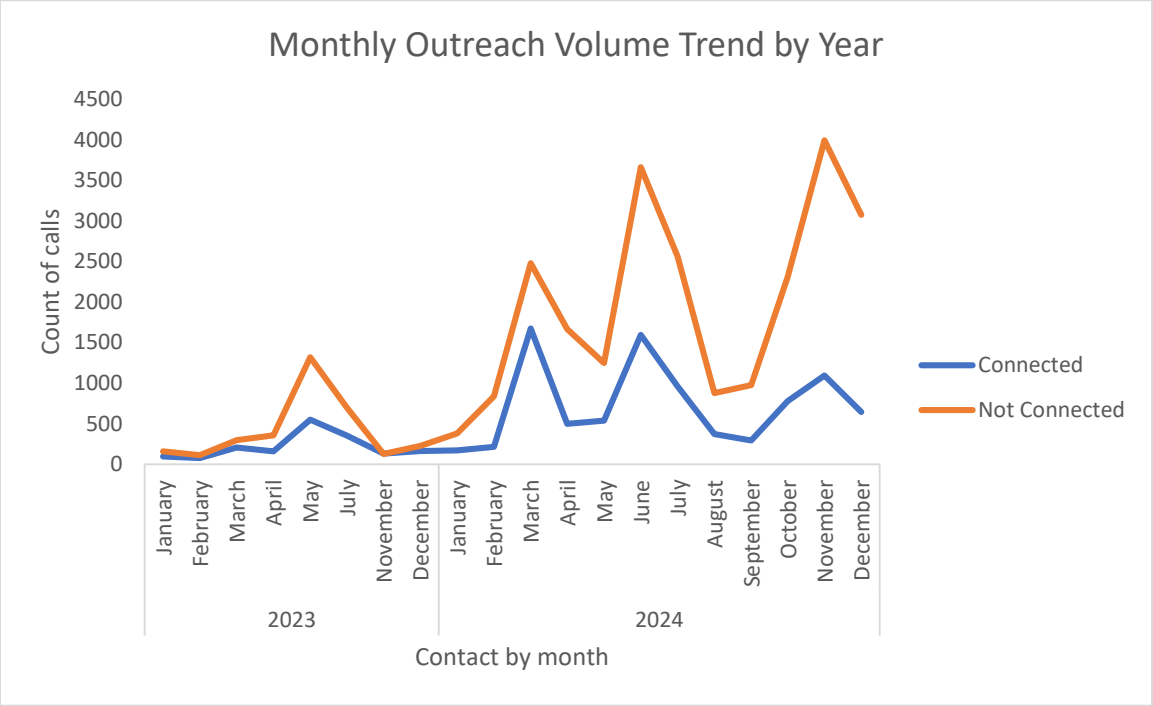
Outreach activity is heavily concentrated during the middle of the week, with the highest call volumes occurring from Monday through Thursday. Both connected and not connected calls peak midweek, particularly on Wednesday, before declining sharply toward the weekend. Saturday records the lowest outreach volume and the fewest successful connections, while Sunday also shows relatively limited activity compared to weekdays.

This pattern suggests that outreach efforts are aligned with traditional working days, when applicants are more reachable and call volumes are maximized. However, the sharp drop in both volume and connections toward the weekend indicates potential underutilization of alternative engagement windows. Fine-tuning outreach schedules by reinforcing high-performing midweek periods while selectively testing weekend strategies could improve overall reachability and contribute to achieving the targeted 15% improvement in enrollment outcomes without increasing total call volume.

Insight 5: Increased Outreach Volume Has Not Translated into Better Outcomes

Outreach volume increased more than sixfold from 2023 to 2024. Despite this increase, the proportion of "Not Connected" outcomes remained consistently high across both years. Outreach activity increased substantially from 2023 to 2024, with significantly higher call volumes recorded across most months in 2024. In 2023, outreach levels were relatively low and uneven, with modest peaks around mid-year and a general decline toward year-end. In contrast, 2024 shows pronounced spikes in outreach volume, particularly around March, June, and October, indicating a major scale-up of campaign activity. Despite this increase in volume, not connected calls continue to dominate across both years.

This trend indicates that while outreach efforts expanded aggressively in 2024, higher call volumes did not translate into proportional improvements in successful connections. The persistence of high non-connection levels suggests structural issues such as reachability, timing, or data quality rather than insufficient effort. To support the goal of improving enrollment outcomes by at least 15%, future campaigns should focus on improving the effectiveness of outreach during peak months through better timing, data validation, and targeting—rather than further increasing call volume alone.



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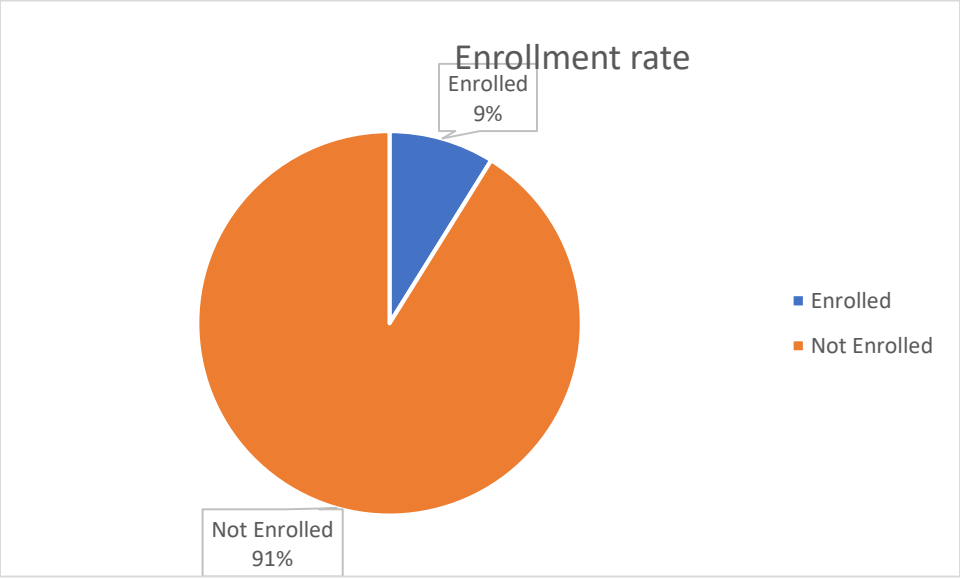
Insight 6: Enrollment and Connection Patterns Differ by Applicant Status

Applicants who eventually enroll show lower connection rates than those who do not enroll, while enrollment outcomes remain heavily concentrated in specific countries. Enrollment outcomes are heavily skewed, with only 9% of applicants ultimately enrolling while 91% do not convert. This indicates that the majority of outreach activity is occurring at the top of the funnel, but only a small fraction translates into final enrollment.

This low enrollment rate highlights a significant conversion gap between outreach and successful enrollment. While outreach volume may be high, the effectiveness of engagement and follow-up appears limited. To achieve the target of improving enrollment outcomes by at least 15%, efforts should focus on strengthening mid- and late-funnel conversion drivers, such as earlier engagement of high-potential applicants, improved follow-up strategies after successful connections, and closer alignment between outreach timing, messaging, and applicant readiness.

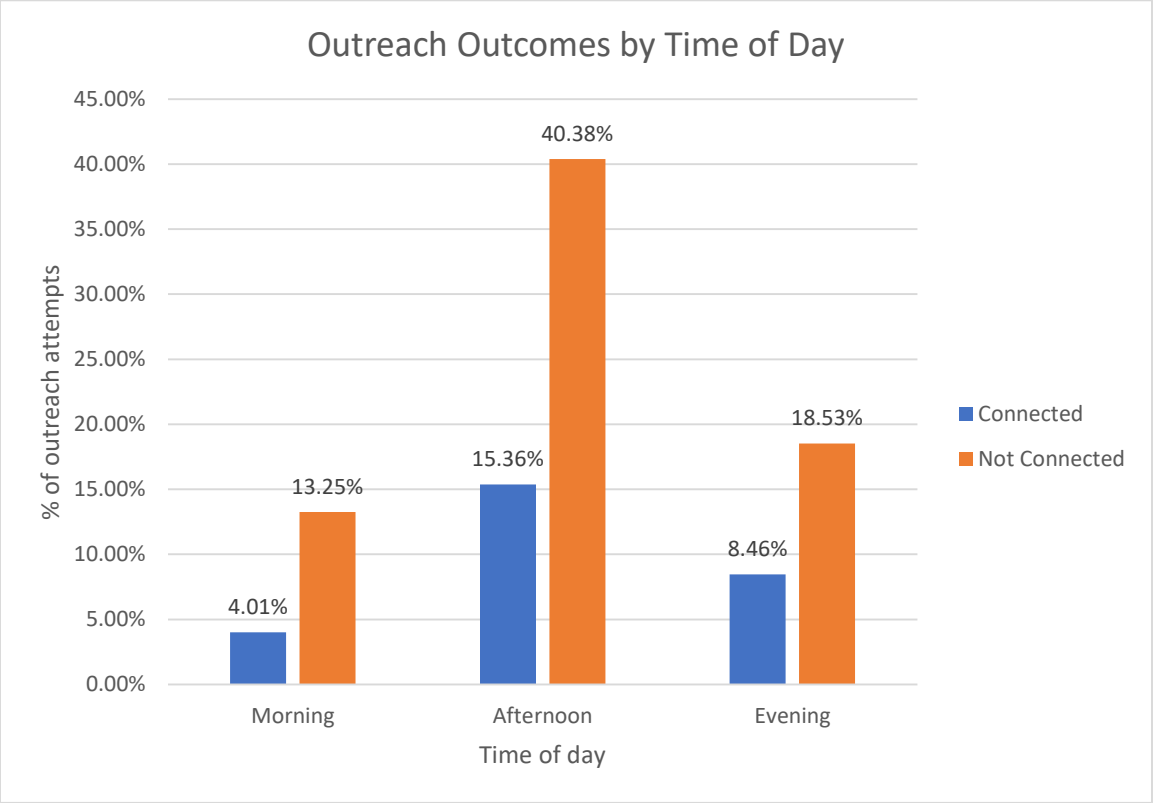


Insight 7 – Evening Calls Outperform Despite Lower Volume

Evening outreach, although representing a smaller share of total call activity, achieves the highest connection rates compared to afternoon and early-day periods. This indicates that applicants are more reachable later in the day, even when overall outreach volume is lower.

This is pertinent because current scheduling appears to prioritize call volume rather than call effectiveness. Concentrating effort in high-volume but lower-performing time windows limits overall connection success and reduces the efficiency of outreach operations.

Reallocating a portion of outreach capacity from afternoon hours to evening periods presents a low-cost opportunity to improve connection rates and overall outreach performance without increasing total call volume. This adjustment directly supports the objective of improving enrollment outcomes by at least 15%.



Conclusion

This exploratory data analysis highlights that Excellerate's primary challenge is not insufficient outreach effort, but limited effectiveness in converting outreach activity into meaningful applicant engagement and enrollment. Across countries, time periods, and years, non-connection outcomes dominate despite substantial increases in outreach volume. The findings consistently show that improvements in reachability—through better timing, data quality, and targeted allocation of outreach resources—offer greater potential for achieving the campaign objective of improving enrollment outcomes by at least 15% than further scaling call volume alone.

Addressing these structural inefficiencies will be critical for translating outreach activity into measurable enrollment gains.

Recommendations

1. **Shift Focus from Volume to Connection Quality**
   Prioritise improving connection rates rather than increasing outreach volume, as higher call volumes alone have not translated into better enrollment outcomes.
2. **Optimise Outreach Timing**
   Concentrate outreach efforts in higher-performing periods, particularly evening hours, and reduce activity during consistently low-yield time windows.
3. **Strengthen Contact Data Quality**
   Implement validation and cleaning of applicant contact information to reduce wasted outreach attempts caused by invalid or unreachable records.
4. **Adopt Targeted, Country-Specific Strategies**
   Maintain scale in high-volume markets while improving reachability, and selectively expand efforts in countries demonstrating higher connectivity and engagement efficiency.
5. **Improve Funnel Conversion After Connection**
   Enhance follow-up and engagement strategies after successful connections to reduce drop-off between outreach and final enrollment.